Final Presentation Outline and Stage Report

Project direction: TabPFN/ICL-FM reproduction + structure-aware feature extension

Last updated: 2026-06-01

Deliverable Scope

Final deliverables:

- A runnable notebook: `notebooks/matbench_tabpfn_official_folds.ipynb`
- A 15-20 min presentation based on the outline below

Reference materials:

- Main paper: `ref_lab_notebook/s41524-026-02089-8_reference.pdf`
- Supplementary file: `ref_lab_notebook/41524_2026_2089_MOESM1_ESM.pdf`
- Course notebooks:
  - `ref_lab_notebook/Lab1_ml_intro_clean.ipynb`
  - `ref_lab_notebook/Lab2_Featurization.ipynb`
  - `ref_lab_notebook/Lab3A_SignLanguage_NN.ipynb`
  - `ref_lab_notebook/Lab3B_SignLanguage_CNN.ipynb`

Current result source:

- Metrics: `results/metrics/model_summary.csv`
- Summary tables: `results/tables/`
- Figures: `results/figures/`

Figure assets copied for the presentation:

- `docs/presentation_assets/03_tabpfn_vs_best_baseline.png`
- `docs/presentation_assets/05_structure_feature_branch_comparison.png`
- `docs/presentation_assets/appendix_01_model_mae_comparison.png`
- `docs/presentation_assets/appendix_02_fold_mae_distribution.png`
- `docs/presentation_assets/appendix_04_tabpfn_parity.png`

Stage Report

What This Project Does

This project evaluates whether a TabPFN-style in-context tabular foundation model can improve small-data materials property prediction on selected Matbench regression tasks. It follows the reference paper's broad ICL-FM idea: convert materials into tabular descriptors, provide train examples as context, and predict held-out Matbench official folds using TabPFN.

The project currently evaluates four Matbench tasks:

Task	Target	Input type	Unit
`matbench_steels`	steel yield strength	composition	MPa
`matbench_expt_gap`	experimental band gap	composition	eV
`matbench_jdft2d`	exfoliation energy	structure	meV/atom
`matbench_phonons`	highest optical phonon peak	structure	cm^-1

Models evaluated:

- Dummy mean baseline
- RidgeCV
- Random forest
- Extra trees
- HistGradientBoosting
- TabPFN
- Post-hoc TabPFN + ExtraTrees 50/50 ensemble
- Inner-validation tuned TabPFN + ExtraTrees ensemble for selected task-feature branches

Feature branches:

- Magpie composition features
- Structure-aware extensions for structure-input tasks:
  - density
  - packing
  - symmetry
  - structural complexity
  - selected combinations
  - all structure descriptors

Evaluation controls:

- Official Matbench five-fold splits
- MAE as primary metric
- R2 as secondary metric
- Fold-level metrics and predictions saved
- TabPFN compared against the strongest non-TabPFN baseline
- No official test-fold tuning for the final model claims

Relationship To The Reference Paper

The reference paper proposes ICL-FM for materials property prediction. It combines TabPFN with tabular materials representations such as Magpie, MagpieEX, and graph-derived ALIGNN/CGCNN embeddings.

This project is a staged reproduction and extension:

- Reproduces the TabPFN/ICL-FM-style workflow using Matbench official folds.
- Uses Magpie and matminer structure descriptors rather than the paper's ALIGNN/CGCNN embeddings.
- Extends the baseline through lightweight structure-aware descriptors.
- Adds automatic summary, error analysis, and ensemble diagnostics.

Current limitation:

- This stage does not reproduce the paper's ALIGNN/CGCNN embedding pipeline.
- Therefore, structure-aware comparisons to FMEF/ALIGNN numbers should be presented as contextual comparison, not exact same-input reproduction.

Connection To The Four Course Notebooks

Use these links to make the final story coherent:

Lab notebook	Reused idea in this project
Lab1 ML intro	Train/test evaluation, regression metrics, materials property prediction setup
Lab2 Featurization	Magpie/matminer featurization, Matbench-style benchmarking, PCA/feature thinking
Lab3A Dense NN	Model complexity, validation behavior, GPU workflow, overfitting/generalization
Lab3B CNN	Motivation for architecture changes and representation learning, while noting that this project uses tabular materials descriptors rather than image CNNs

The strongest continuity is Lab2. The project can be framed as: Lab2 introduced materials featurization and Matbench; the reference paper uses TabPFN as an in-context model on such features; this project tests that idea and extends the featurization for structure-dependent tasks.

Quantitative Published-Paper Comparison

Sources:

- Main paper Table 1: composition-based Matbench results for `jdft2d` and `phonons`.
- Main paper Table 2/Table 4: structure-based Matbench results for `jdft2d` and `phonons`.
- Supplementary Table S13: composition-only Matbench results for `steels` and `expt_gap`.

Lower MAE is better.

Task	Representation	Our best MAE	Published comparison	Difference	Interpretation
`matbench_steels`	Magpie composition	87.004 MPa	ICL-FM 83.605 MPa	+4.07%	Slightly worse than paper ICL-FM
`matbench_steels`	Magpie composition	87.004 MPa	TPOT-Mat 79.947 MPa	+8.83%	Worse than composition-only leaderboard best
`matbench_expt_gap`	Magpie composition	0.3688 eV	ICL-FM 0.3142 eV	+17.38%	Clear gap to paper ICL-FM
`matbench_expt_gap`	Magpie composition	0.3688 eV	Darwin 0.2865 eV	+28.73%	Clear gap to leaderboard best
`matbench_jdft2d`	Magpie composition proxy	46.130 meV/atom	FM-Magpie 44.736 meV/atom	+3.12%	Close to paper composition-only FM
`matbench_phonons`	Magpie composition proxy	36.853 cm^-1	FM-MagpieEX 41.948 cm^-1	-12.15%	Numerically better than reported FM-MagpieEX, but treat as a strong result that should be rerun/verified before making it a headline claim
`matbench_jdft2d`	Magpie + structure descriptors	34.396 meV/atom	FMEF 40.880 meV/atom	-15.86%	Numerically lower than paper FMEF, but not exact same representation; do not present as beating the paper's full structure method
`matbench_jdft2d`	Magpie + structure descriptors	34.396 meV/atom	SOTA MODNet 33.192 meV/atom	+3.63%	Near but not better than SOTA
`matbench_phonons`	Magpie + structure descriptors	30.631 cm^-1	FMEF 29.753 cm^-1	+2.95%	Close to paper FMEF contextual number
`matbench_phonons`	Magpie + structure descriptors	30.631 cm^-1	SOTA MegNet 28.761 cm^-1	+6.50%	Competitive but not SOTA

Important caveat for slides:

The `jdft2d` and `phonons` structure-aware rows are not exact same-representation reproduction, because this project uses lightweight matminer structure descriptors, while the paper also reports graph-derived ALIGNN/CGCNN embeddings. Present this as a staged extension rather than a full graph-embedding reproduction. Also avoid saying the project "beats the published paper" overall: `jdft2d` remains worse than the paper's published SOTA MODNet number, and `phonons` remains worse than the paper's published structure-aware SOTA MegNet number.

Main Scientific Conclusions

1. TabPFN is competitive on selected small-data Matbench tasks, but it does not uniformly beat the published paper or leaderboard numbers.
2. The strongest result is the structure-aware featurization extension.
3. On structure-input tasks, composition-only features are insufficient:
   - `matbench_jdft2d`: best structure branch improves over composition proxy by 25.44% MAE.
   - `matbench_phonons`: best structure branch improves over composition proxy by 16.88% MAE.
4. Simple TabPFN + ExtraTrees ensembling is not a reliable improvement.
5. The project is now more defensible as a feature-representation study than as a claim that TabPFN alone is always better.

Recommended Main Figures

Figure 1 - TabPFN Relative To Classical Baselines

Use this to show where TabPFN beats or loses to the strongest non-TabPFN baseline.

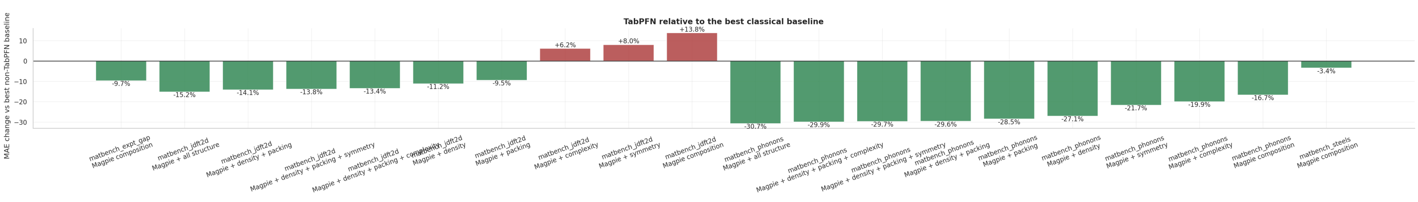

Figure 1: TabPFN vs best baseline

Speaker point:

Green bars mean TabPFN improves MAE relative to the strongest classical baseline. Red bars mean it does not. The figure supports a nuanced claim: TabPFN is often strong, especially with useful representations, but not automatically best for every feature branch.

Figure 2 - Structure Feature Branch Ablation

Use this as the central scientific figure.

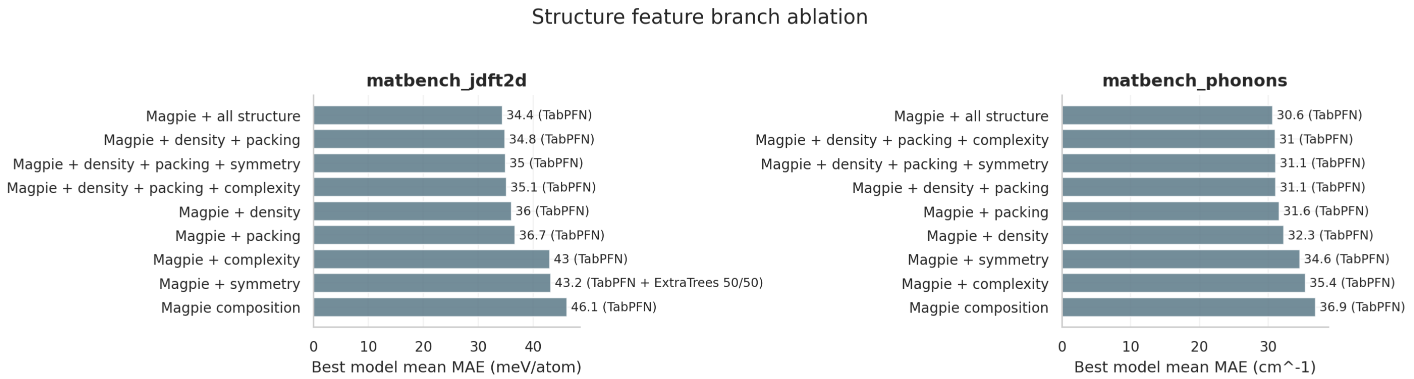

Figure 2: Structure feature branch ablation

Speaker point:

Adding physically meaningful structure descriptors gives a large improvement on structure-dependent targets. The best branch is Magpie plus all structure descriptors for both `jdft2d` and `phonons`. Density and packing explain much of the gain; symmetry alone is weaker.

15-20 Min Slide Outline With Speaker Notes

Target length: 13 slides, about 16-18 minutes.

Slide 1 - Title

Title:

TabPFN/ICL-FM for Small-Data Materials Property Prediction

Subtitle:

Reproduction-style Matbench study with structure-aware feature extension

Timing: 0.5-1 min

Speaker notes:

Introduce the project as a staged reproduction and extension of the ICL-FM paper. The goal is not only to run TabPFN, but to test whether the feature representation changes the scientific conclusion on small materials datasets.

Slide 2 - Motivation

Main points:

- Materials datasets are often small, expensive, and heterogeneous.
- Conventional ML needs careful feature engineering and validation.
- Foundation models like TabPFN promise fast in-context prediction without task-specific training.

Timing: 1-1.5 min

Speaker notes:

Connect this to the course arc. Lab1 introduced supervised property prediction. Lab2 showed that descriptors strongly affect performance. The reference paper asks whether a pretrained tabular foundation model can make small-data prediction easier and stronger.

Slide 3 - Reference Paper

Main points:

- Paper: In Context Learning Foundation Models for Materials Property Prediction with Small Datasets.
- Core method: ICL-FM = TabPFN + materials descriptors.
- Representations: Magpie, MagpieEX, and graph-derived ALIGNN/CGCNN embeddings.
- Benchmark: Matbench official folds.

Timing: 1.5 min

Speaker notes:

Explain that TabPFN is the prediction engine. The paper's novelty is using it as an in-context model over materials features. Emphasize that the paper reports both composition-based and structure-based results, which lets us compare our numbers quantitatively.

Slide 4 - Course Notebook Continuity

Main points:

- Lab1: regression workflow, train/test split, metrics.
- Lab2: featurization with Magpie/matminer and Matbench thinking.
- Lab3A/B: model complexity, GPU workflow, validation and generalization.

Timing: 1 min

Speaker notes:

This project is not isolated from the labs. It extends Lab2 most directly: instead of only using classical ML on fixed features, we use TabPFN and test whether adding structure-aware descriptors improves the result.

Slide 5 - Research Questions

Main questions:

1. Can we reproduce a similar TabPFN/ICL-FM-style workflow on Matbench official folds?
2. How close are our results to the published paper numbers?
3. Does structure-aware featurization improve over composition-only Magpie features?
4. Does a simple TabPFN + ExtraTrees ensemble improve the final model?

Timing: 1 min

Speaker notes:

Frame the project as two parts: reproduction-style benchmarking and extension. The extension is the main contribution, because we do not implement the paper's ALIGNN/CGCNN embedding pipeline in this stage.

Slide 6 - Data And Tasks

Main points:

Task	Samples	Target	Unit
`matbench_steels`	312	yield strength	MPa
`matbench_jdft2d`	636	exfoliation energy	meV/atom
`matbench_phonons`	1,265	phonon peak frequency	cm^-1
`matbench_expt_gap`	4,604	experimental band gap	eV

Timing: 1-1.5 min

Speaker notes:

These tasks span composition-only and structure-input problems. The small sizes are important: they are where an in-context model like TabPFN should be useful.

Slide 7 - Methods

Main points:

- Official Matbench five-fold splits.
- Same features used for all models within each task-feature branch.
- Baselines: Dummy, RidgeCV, Random Forest, ExtraTrees, HistGradientBoosting.
- Main model: TabPFN.
- Extension: structure descriptors from matminer.
- Diagnostics: fixed ensemble, inner-validation tuned ensemble, top-error analysis.

Timing: 1.5-2 min

Speaker notes:

Stress that the evaluation protocol avoids test-fold tuning for final claims. The ensemble weight scan is diagnostic only. The inner-validation tuned ensemble is included to check whether a leakage-free ensemble helps.

Slide 8 - Quantitative Comparison To Published Numbers

Main table:

Task	Our best	Published comparison	Result
`steels`	87.004 MPa	ICL-FM 83.605 MPa	4.07% worse
`expt_gap`	0.3688 eV	ICL-FM 0.3142 eV	17.38% worse
`jdft2d` composition	46.130 meV/atom	FM-Magpie 44.736	3.12% worse
`phonons` composition	36.853 cm^-1	FM-MagpieEX 41.948	12.15% lower; verify before headline claim
`jdft2d` structure	34.396 meV/atom	SOTA 33.192	3.63% worse
`phonons` structure	30.631 cm^-1	SOTA 28.761	6.50% worse

Timing: 2 min

Speaker notes:

This slide should be presented carefully. The project does not beat the paper overall. It is close on `jdft2d`, numerically strong on `phonons` composition, and competitive on structure tasks. The biggest weakness is `expt_gap`, where the paper's ICL-FM and Darwin are clearly better.

Slide 9 - TabPFN Versus Classical Baselines

Figure:

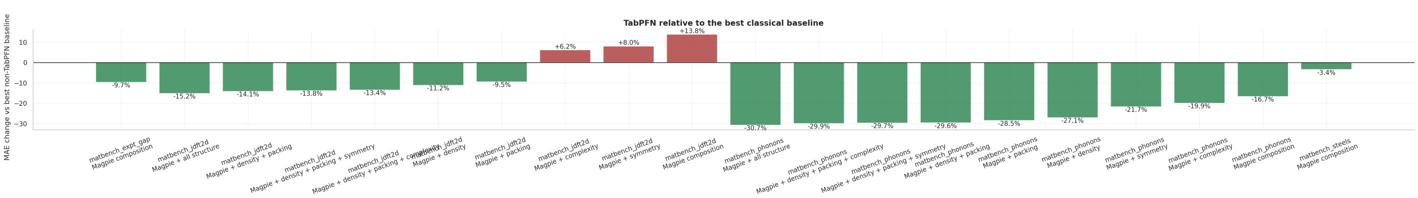

Figure 3: TabPFN vs best baseline

Timing: 1.5 min

Speaker notes:

Use this figure to explain internal comparison within our experiment. TabPFN beats strong classical baselines in many feature branches, especially on phonons and structure-enhanced settings. However, red bars show that TabPFN does not automatically win when the feature branch is weak.

Slide 10 - Structure-Aware Feature Extension

Figure:

Timing: 2 min

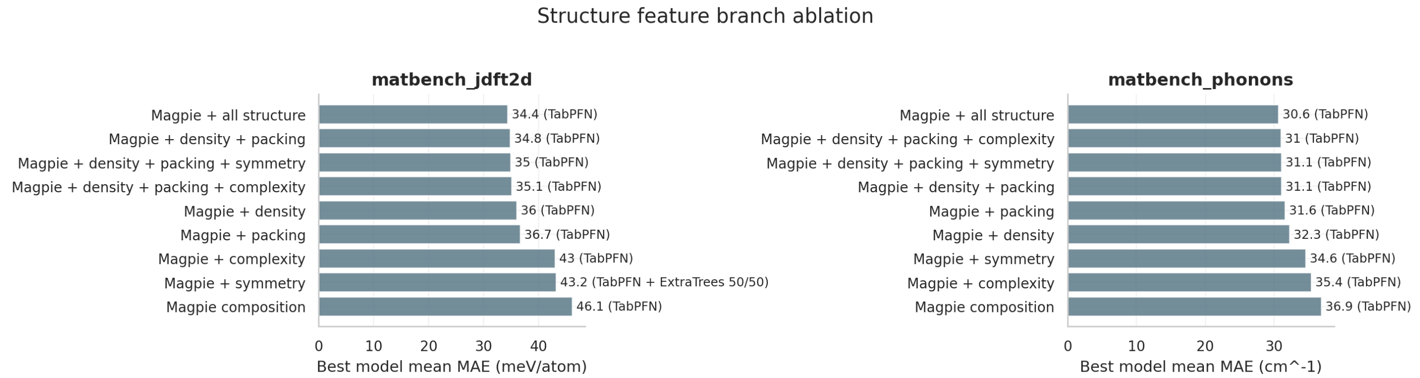

Figure 4: Structure feature branch ablation

Speaker notes:

This is the central project contribution. The structure-aware branch moves `jdft2d` from 46.13 to 34.40 MAE and `phonons` from 36.85 to 30.63 MAE. The result supports the physical expectation that structure matters for exfoliation energy and phonon frequencies.

Slide 11 - Ensemble And Error Analysis

Main points:

- Fixed 50/50 TabPFN + ExtraTrees helps only slightly on `steels`.
- Inner-validation tuned ensemble does not reliably beat standalone TabPFN.
- `expt_gap` has difficult outliers and a catastrophic RidgeCV fold, indicating linear extrapolation failure.
- Top-error tables show errors concentrate in extreme target regimes.

Timing: 1.5 min

Speaker notes:

This is a useful negative result. The improvement does not come from a trivial ensemble. It comes mainly from representation. Also explain that the RidgeCV outlier is not a GPU training issue; it is a fold-specific failure of a linear baseline on Magpie features.

Slide 12 - Limitations

Main points:

- This stage does not include paper's ALIGNN/CGCNN graph embeddings.
- The published paper may use different TabPFN versions and preprocessing choices.
- Some current figures are too dense for final slides and should be simplified in the final deck.
- `expt_gap` remains below paper ICL-FM and leaderboard best.

Timing: 1 min

Speaker notes:

Be explicit about what was not done. This makes the project more credible. The correct claim is not "we fully reproduce the paper"; it is "we reproduced the TabPFN/ICL-FM-style pipeline and extended featurization with lightweight structure descriptors."

Slide 13 - Conclusions And Next Steps

Main conclusions:

- TabPFN is a strong small-data materials predictor but not universally SOTA.
- Feature representation is the decisive factor.
- Lightweight structure descriptors substantially improve structure-dependent tasks.
- The project is a coherent staged extension of the reference paper and Lab2 featurization work.

Next steps:

- Clean final slides with compact tables and two main figures.
- Add a short final notebook cell summarizing published-paper comparison.
- Optional future extension: implement ALIGNN/CGCNN embeddings if more time is available.

Timing: 1-1.5 min

Speaker notes:

End with the balanced claim: the project shows that in-context TabPFN can be effective, but the most scientifically important result is that physically meaningful structure-aware features are necessary for structure-property prediction.

Appendix Figures

These figures are copied for backup or appendix slides. They are useful for Q&A, but too dense for the main 15-20 min presentation.

Appendix Figure A - Full Model MAE Comparison

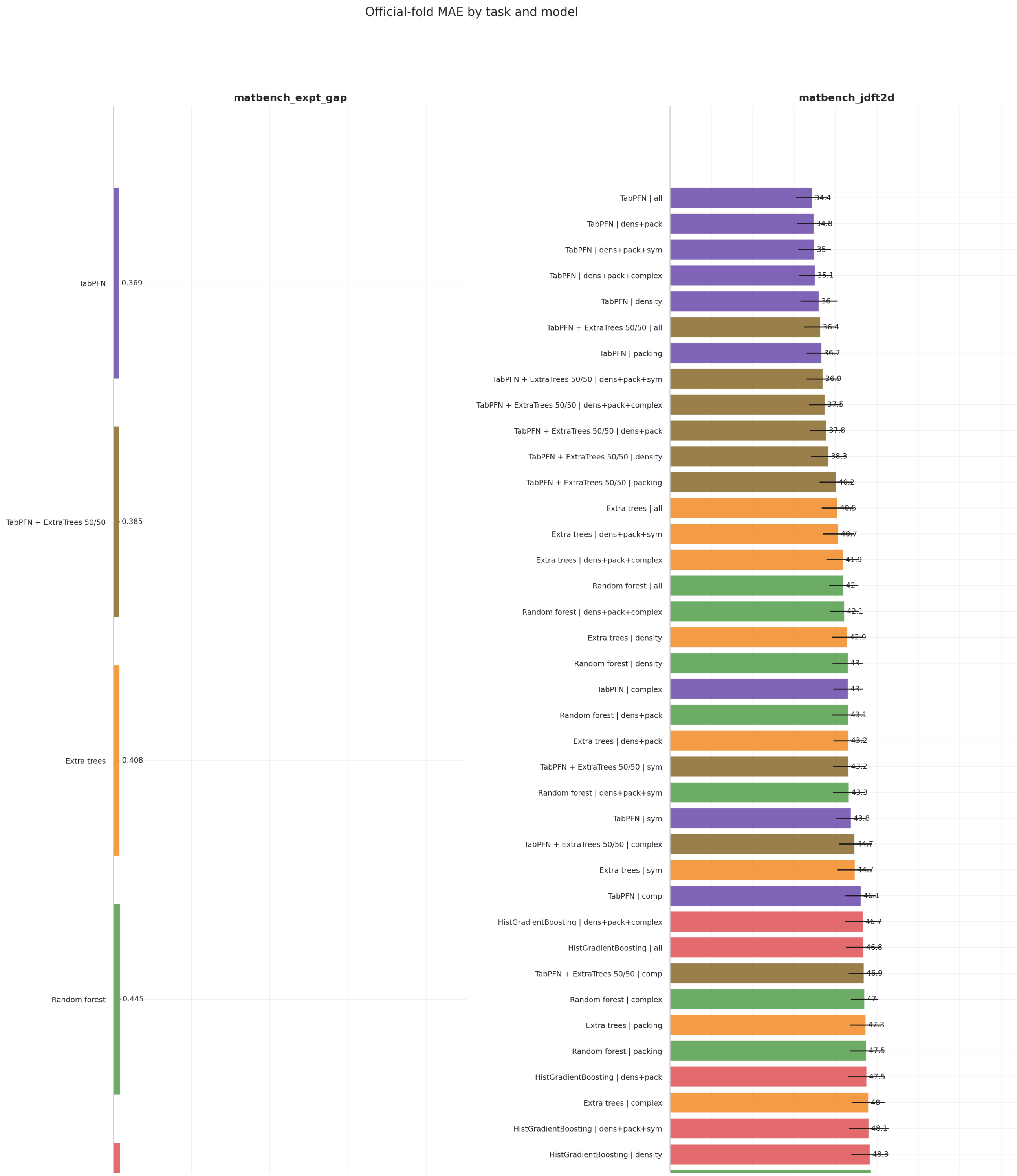

Figure 5: Full model MAE comparison - part 1 of 3

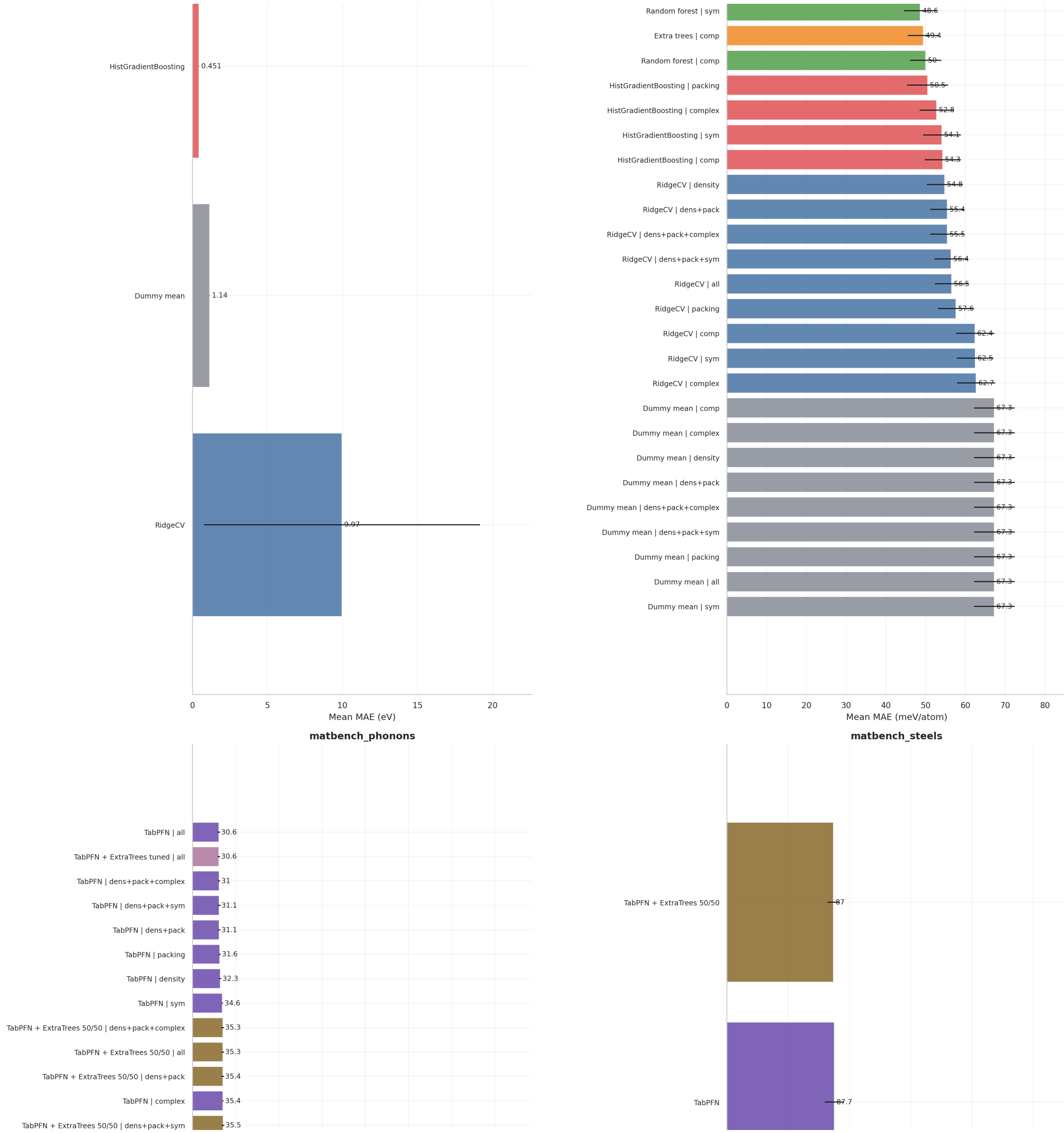

Figure 6: Full model MAE comparison - part 2 of 3

Appendix Figure B - Fold-Level MAE Dispersion

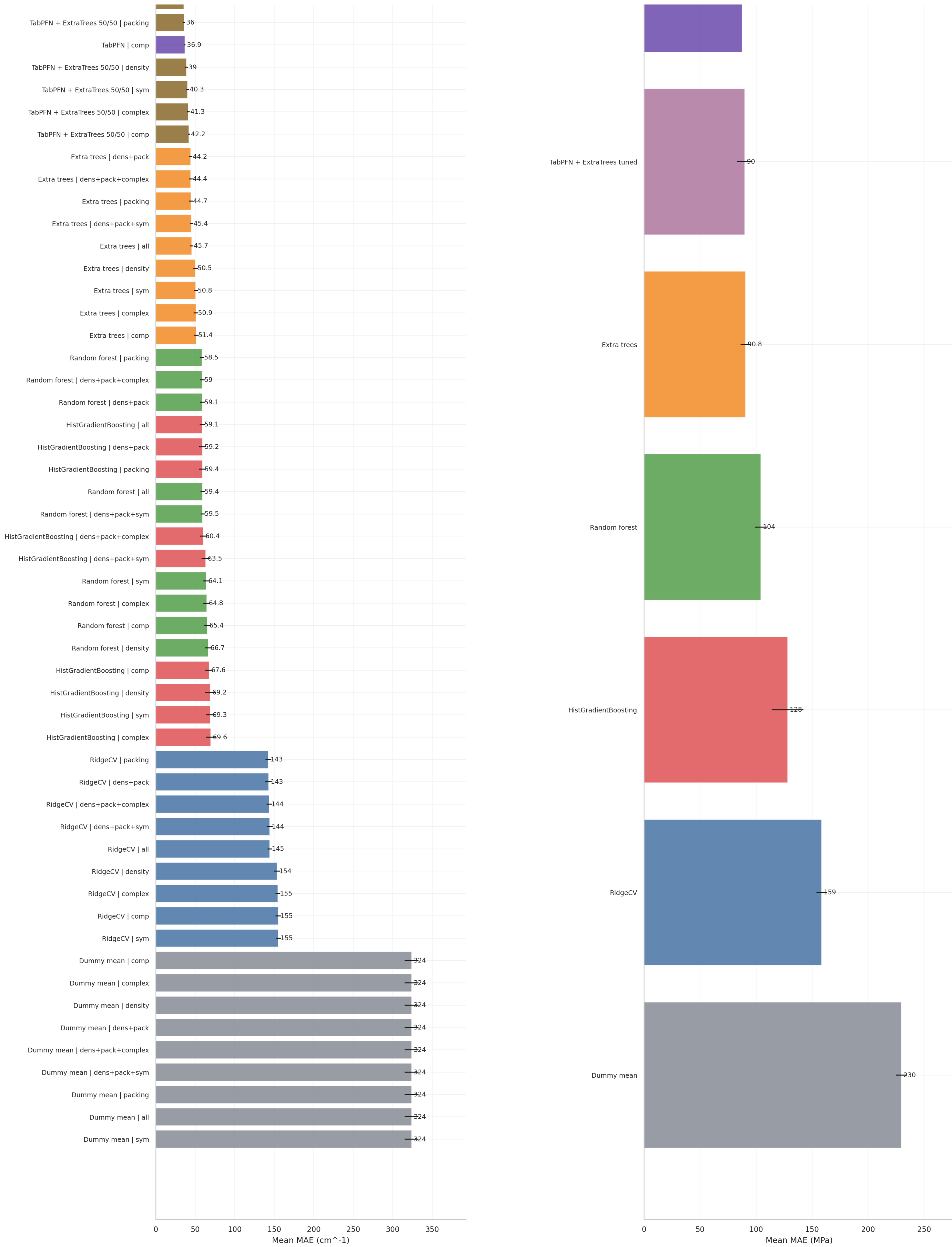

Figure 7: Full model MAE comparison - part 3 of 3

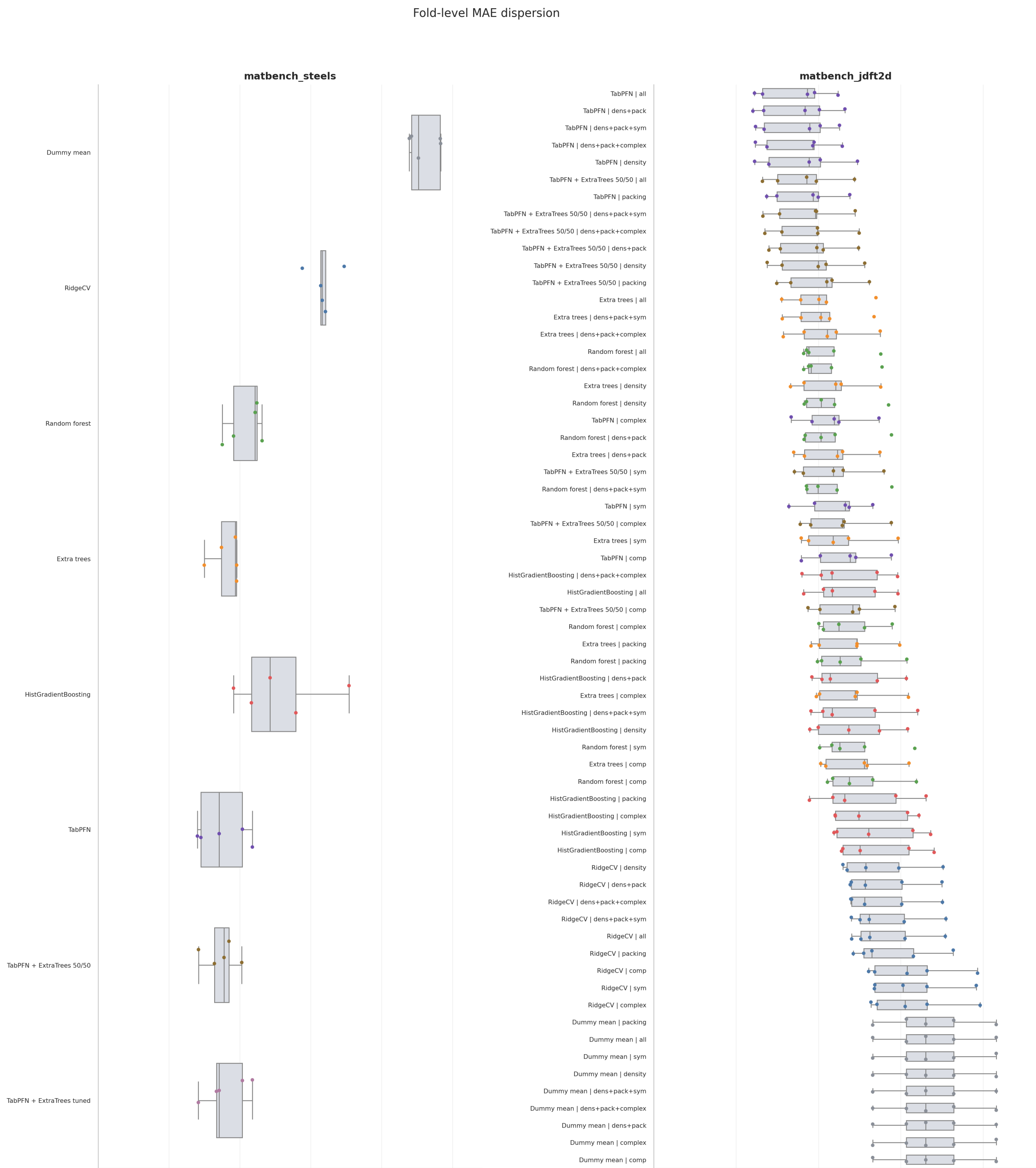

Figure 8: Fold-level MAE dispersion - part 1 of 2

Appendix Figure C - Parity Plots

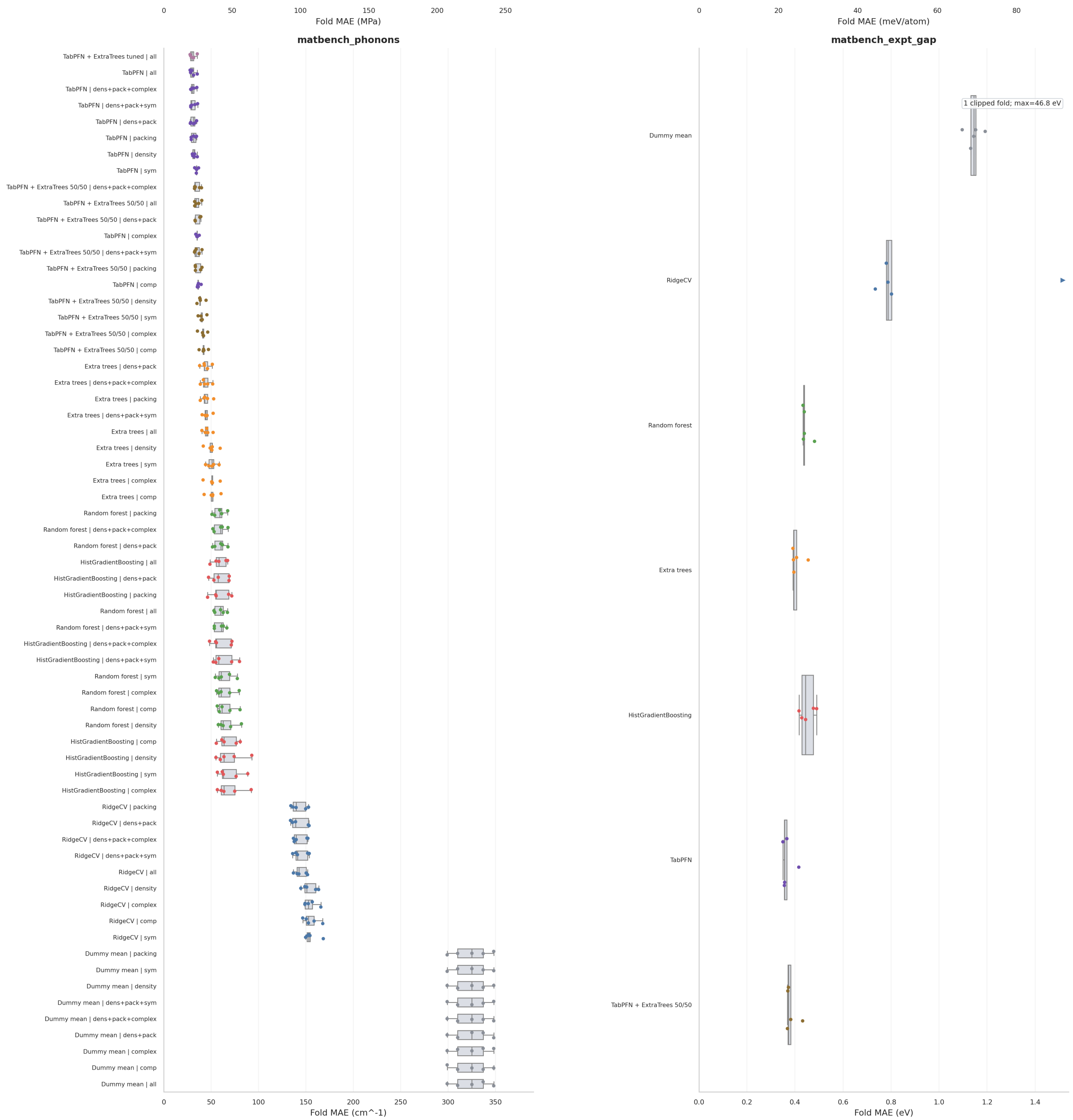

Figure 9: Fold-level MAE dispersion - part 2 of 2

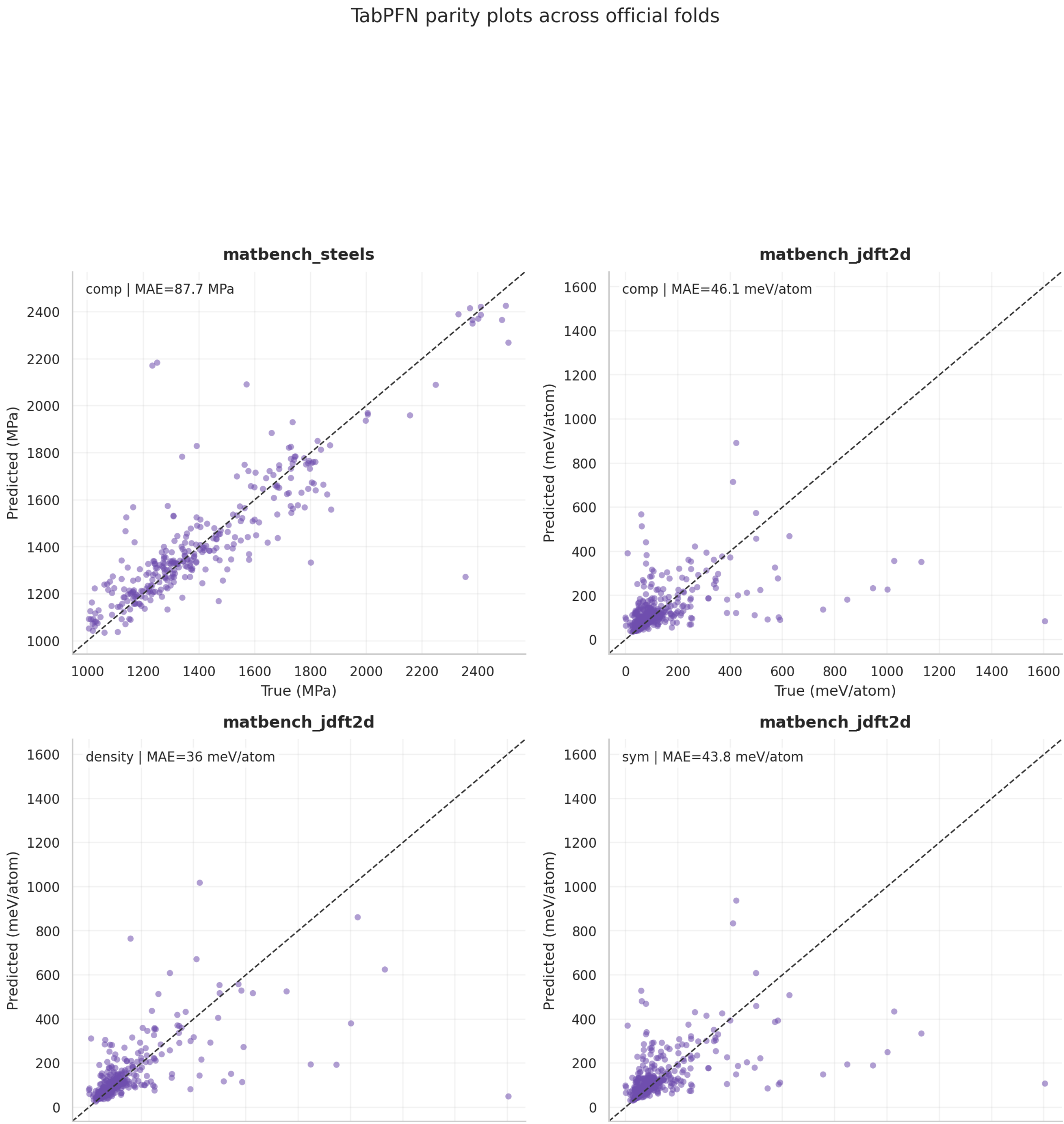

Figure 10: Parity plots - part 1 of 5

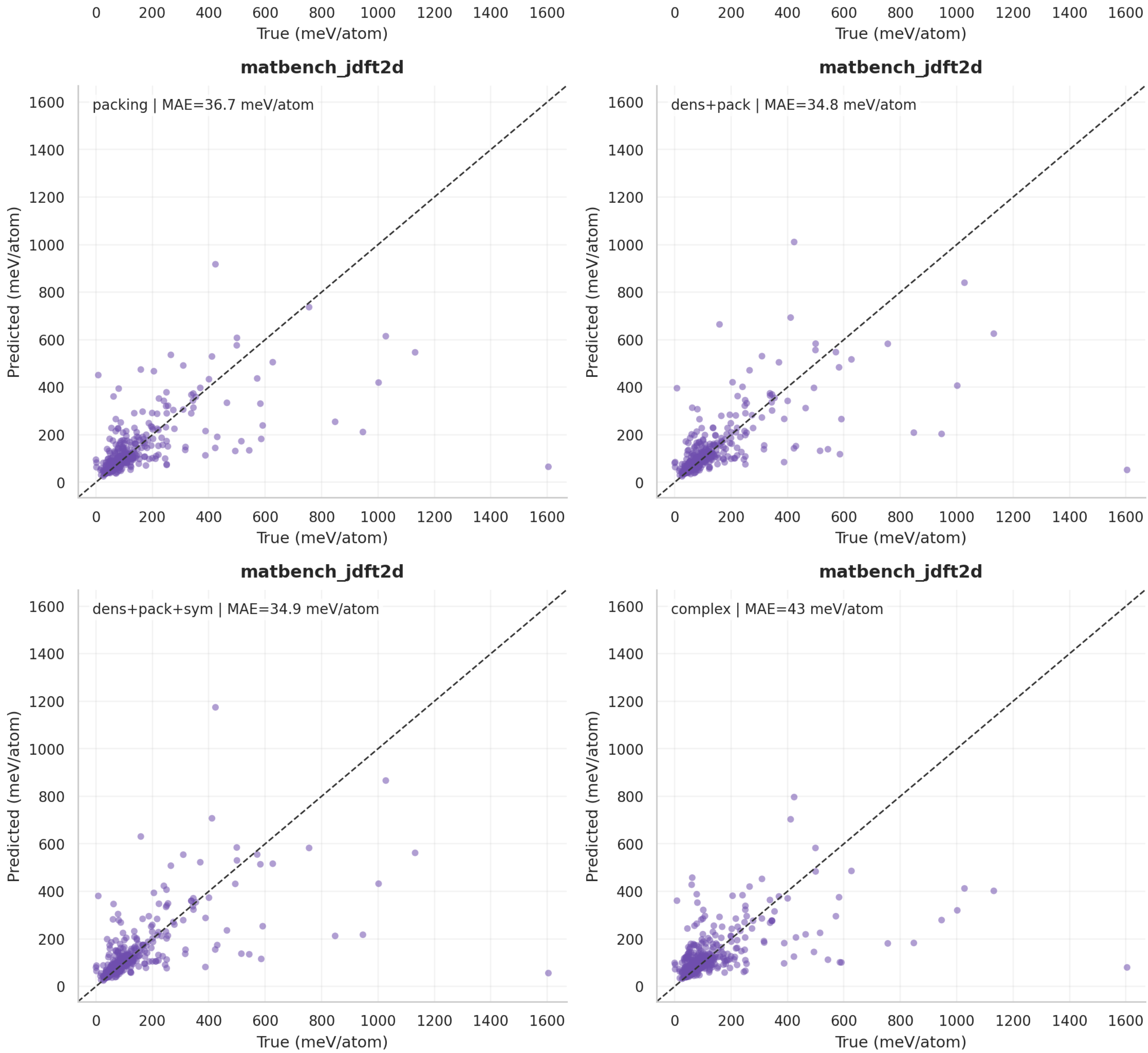

Figure 11: Parity plots - part 2 of 5

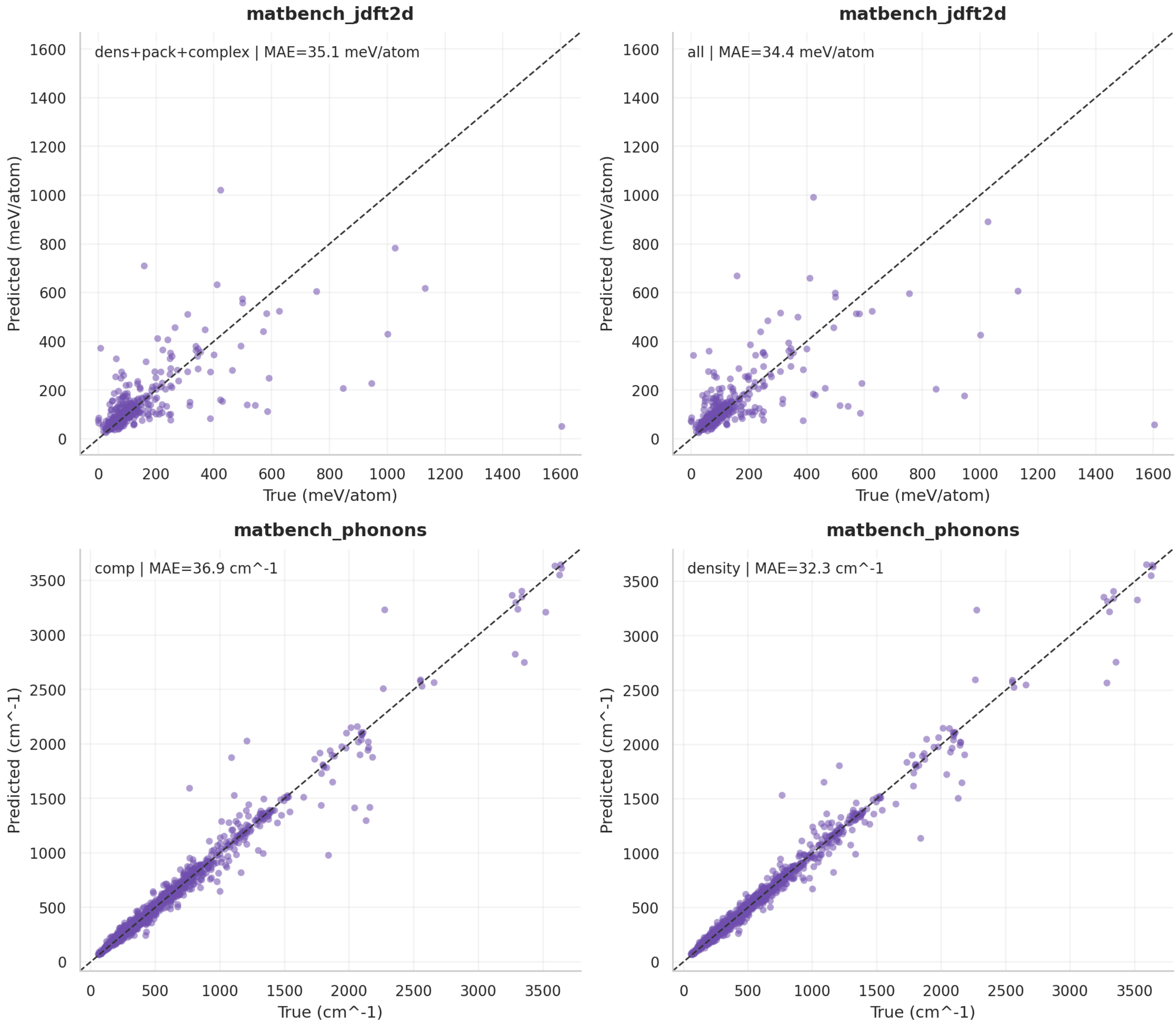

Figure 12: Parity plots - part 3 of 5

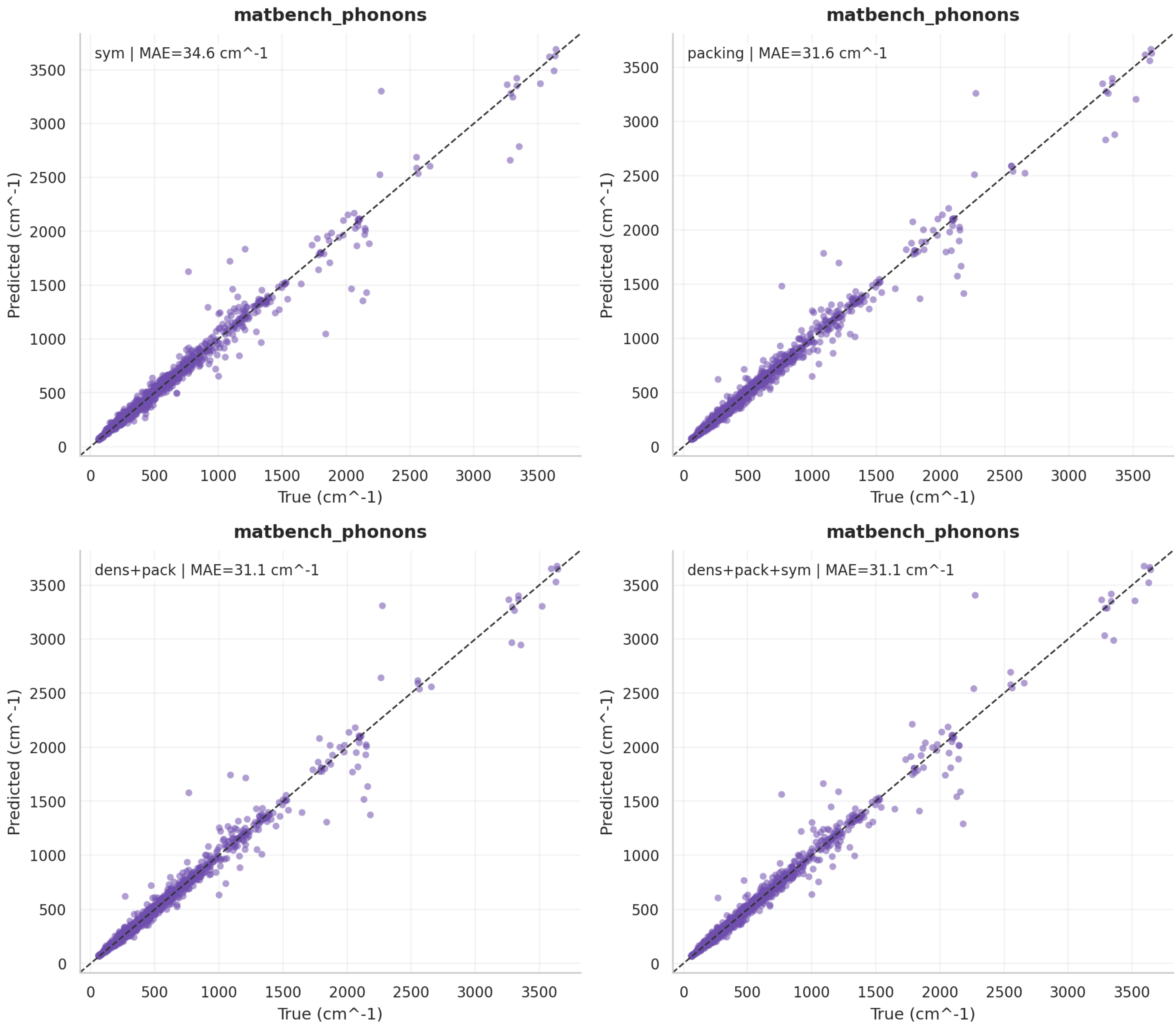

Figure 13: Parity plots - part 4 of 5

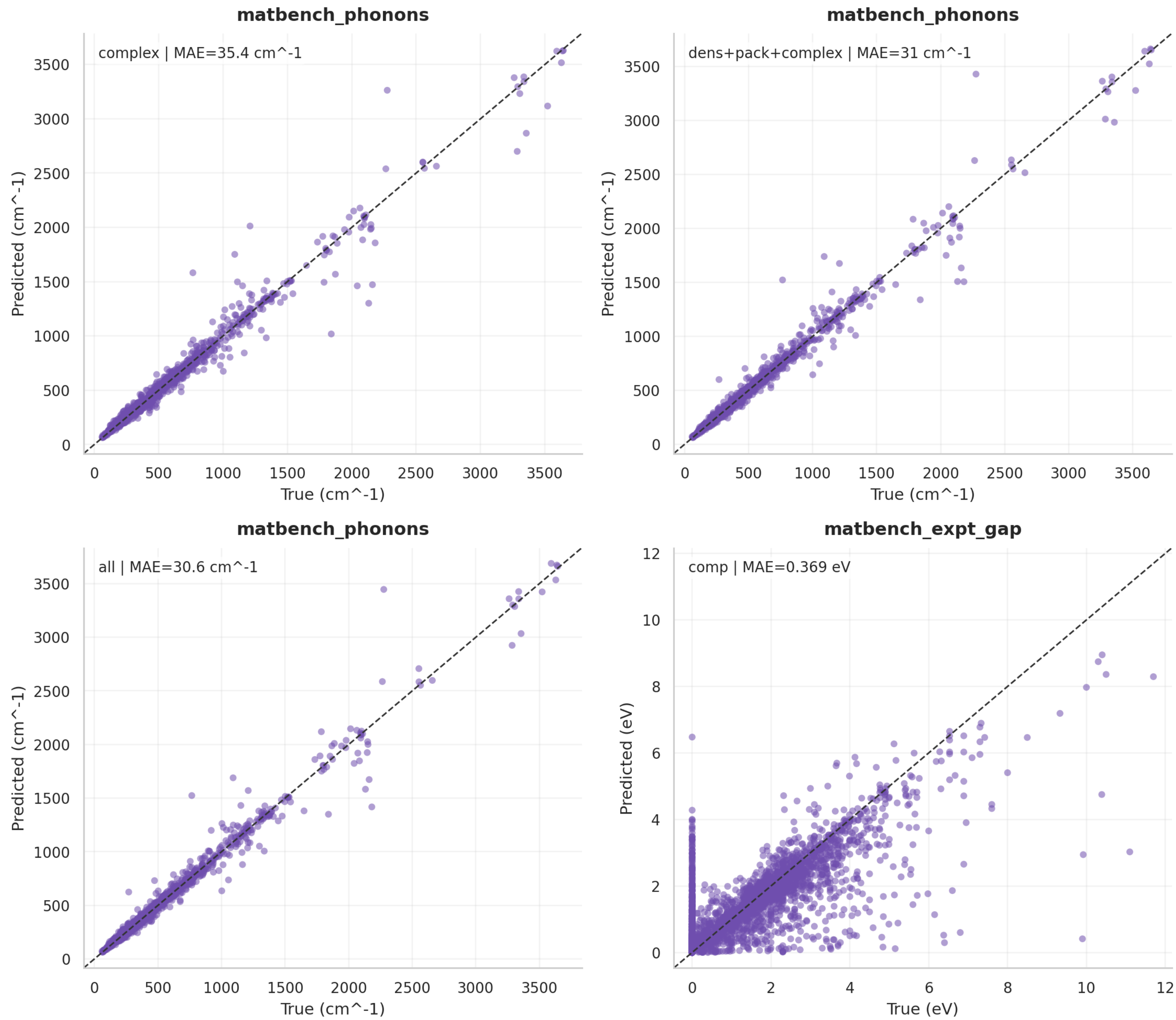

Figure 14: Parity plots - part 5 of 5

Final Deck Construction Notes

Recommended main-deck visuals:

- Use Slide 8 as a compact quantitative table.
- Use Slide 9 for internal TabPFN-vs-baseline comparison.
- Use Slide 10 as the central scientific result.
- Keep full model comparison, fold dispersion, and parity plots in appendix only.

Recommended final claim:

This project provides a staged reproduction of a TabPFN/ICL-FM-style Matbench workflow and extends it with lightweight structure-aware descriptors. The project does not fully reproduce the paper's ALIGNN/CGCNN embedding pipeline, but it shows that structure-aware featurization substantially improves TabPFN performance on structure-dependent materials properties and brings results close to published structure-aware benchmarks.